

Omar A. Mohamed

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EDUCATION

California Polytechnic State University

B.S. in Computer Science

San Luis Obispo, California

Expected Graduation, Jun. 2028

- GPA: 3.6/4.0 | Honors & Awards: President's List, Dean's List (4x)
- Related Coursework: Data Structures, Algorithm Design and Analysis, Computer Organization, Object-Oriented Programming, Systems Programming, Discrete Structures, Linear Algebra, Database Systems, Cybersecurity

CodePath

Web Development (WEB103), AI Engineering (AI201), Technical Interview Prep (TIP102)

Remote

EXPERIENCE

Sandbar Health

Software Engineering Intern

Remote

Jul. 2026 – Aug. 2026

- Architected and shipped a production **AI-powered broker tool**, automating document extraction, renewal analysis, plan comparison, and carrier outreach across 28 API endpoints using **Next.js**, **TypeScript**, and **PostgreSQL**.
- Re-engineered the broker platform **authentication flow**, reducing observed sign-in failures from **80–90% to zero** by rebuilding invite/callback handling and secure multi-org routing using **WorkOS AuthKit**, **OAuth 2.0**, and **PKCE**.
- Built **LLM extraction pipelines** supporting 2–5 plans per insurance document, transforming unstructured quotes and PDFs into structured comparison data using **Claude API**, schema validation, background processing, and cached results.

Algoverse

Machine Learning Researcher

Remote

Mar. 2026 – Jul. 2026

- Improved HumanEval pass@1 from **43.3% to 60.4% (+17.1 pts)** by implementing and evaluating On-Policy Distillation on Qwen2.5-Coder-1.5B using **PyTorch**, **Transformers**, **PEFT**, and **QLoRA**.
- Improved Simple Self-Distillation HumanEval performance from **70/164 to 77/164 (+4.3 pts)** by designing ablation studies that identified noisy self-generated targets and introducing a code-extraction preprocessing pipeline.
- **Co-authored an ML research manuscript** on LLM self-improvement while building reproducible single-GPU training and evaluation pipelines on Colab (T4), including debugging benchmark failures that restored HumanEval results from 0/164 to 72/164.

CS+AI Club

Director of Meetings

San Luis Obispo, California

Meeting Facilitator

Aug. 2026 – Present

Sep. 2025 – May 2026

- Designed and led 3 AI workshops for ~40 students, creating Google Colab demos and coding exercises using **Gemini**, **LangChain**, **ChromaDB**, **pandas**, and **scikit-learn** to teach prompt engineering, RAG, and applied AI workflows.

TECHNICAL PROJECTS

Improv Faceoff | CodeboxHacks 2026 Hackathon Project

Sep. 2026

- Built and deployed a **real-time 1v1 improv training game** in 30 hours on a 3-person team, enabling remote players to match, communicate over live video/audio, complete synchronized 60-second rounds, and receive AI-generated feedback
- Synchronized **multiplayer state** and live media across two remote clients by engineering a **Python/Flask-SocketIO backend** with **Redis**-backed atomic matchmaking, server-authoritative game state, and match-scoped **LiveKit** rooms

New News AI | Kiro Hacks 2026 Hackathon Project

May 2026

- Built and deployed a **multi-agent news platform** to **Vercel** in 13 hours on a 2-person team, enabling users to discover and validate breaking-news stories and explore them through real-time voice conversations
- Enabled autonomous news analysis and context-aware voice discussions by engineering both agents end-to-end, using **GPT-4o-mini**, **OpenAI tool calling**, and **Tavily MCP** for iterative retrieval and validation, and **OpenAI Realtime API** + **WebRTC** to maintain conversational context and trigger dynamic follow-up searches during live discussions

KitchLog | CodePath WEB103 Capstone Project

Jul. 2026

- Built and deployed recipe-import and cooking-plan features for a **full-stack recipe app** on a **4-person team** using **React**, **Node.js**, **Express**, **PostgreSQL**, and **REST APIs**, contributing 2,100 LOC across 11 commits and 3 PRs.

SKILLS

Languages & Frameworks: Python, Java, JavaScript, C++, HTML/CSS, Next.js, React.js, SQL

Tools: Node.js, OpenAI API, LangChain, ChromaDB, Supabase, TensorFlow, PyTorch, OpenCV, scikit-learn, pandas, Git, GitHub